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RESEARCH ARTICLE

The Right to Restrict AI Training

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Abstract
Generative AI systems require vast amounts of training data, much of it scraped from the internet without creators' consent. Critics often characterize this practice as "theft," but such claims require showing that AI training violates creators' property rights. However, this position risks implying that human learning and works of inspiration would count as theft, too. This paper develops two arguments for moral restrictions that apply uniquely to AI training, not to human learning. The first contends that even if creators possess extensive moral property rights over the use of their content, in the human case, these property rights are typically overridden by a countervailing right to freedom of thought. Since AI systems lack such a right, creators' property rights remain intact, requiring AI companies to obtain permission to use their work. The second argument contends that even without such broad property rights, creators have the moral right to restrict which copying technologies may be used to capture their work, allowing them to block web crawlers for AI training while permitting ordinary human browsing. The paper concludes by exploring the policy implications of these moral arguments.

Keywords AI ethics · Training data · Intellectual property · Freedom of speech · Freedom of thought

1 Introduction

Generative artificial intelligence requires reams of data, and its developers have set loose a fleet of bots to trawl the internet for whatever they can catch. Unsurprisingly, much of the material swept up in the dragnet did not end up there with the approval of

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its creators. This has led to widespread accusations of "theft" against AI companies.[1] The accusation: the AI training process exploits content made by others for profit, without compensation or consent. With the click of a button, AI companies appropriate the value others have produced. That looks a lot like theft with extra steps.

While anti-AI sentiment is rampant among creators, their accusations are generally left underspecified. For one thing, it is often unclear what is alleged to have been stolen: the creators' content, the data derived from it, or its value. For another, accusations of theft and exploitation are frequently conflated. In any case, the core complaint is that AI companies are somehow violating creators' rights to control what they create—i.e., their *property rights*. Because exploitation introduces an array of difficulties (e.g., power disparities, coercion, etc.) that I cannot hope to address here, I will frame the matter in terms of ordinary theft. This should not be taken to suggest that AI training could not involve both theft *and* exploitation, but I leave that matter aside.[2]

A successful accusation of theft requires showing that AI training violates property rights. But what sort of property rights might restrict AI training specifically, while leaving human learning untouched? As Shoemaker points out, human artists routinely learn from and incorporate the styles and techniques of others without permission, and society does not view this as theft—even when it causes economic harm.[3] The challenge, then, is to identify a property right that allows creators to block AI training without unduly restricting human creativity. More specifically, I am interested in identifying such a *moral* right rather than a legal one. Insofar as moral reasoning ought to constrain law and policy, it makes sense to begin with the moral question.

This paper explores two dialectical strategies to this end. The first begins by granting, for the sake of argument, that creators possess extensive, patent-like intellectual property rights to block others from using their content, including for learning. Even if this is true, I argue that humans possess a countervailing right to freedom of thought that typically overrides these rights. Since AI systems lack a right to freedom of thought, this justification for bypassing creators' rights does not apply to AI training. If successful, this argument shows how creators could hold extensive intellectual property rights that restrict AI training without unduly limiting human creativity. While this does not establish that creators in fact possess such rights, it removes an objection that would otherwise render a whole family of proposals nonstarters, clearing the way for further exploration of such views.

Still, the ethics of intellectual property is controversial, and many will be skeptical of granting creators such broad intellectual property rights, even in *pro tanto* form. I therefore offer a second, narrower justification for restricting AI training that focuses

[1] See Goetze (2024) for a Lockean argument to this effect. Hayes (2023) provides an extended ethical overview of generative AI and copyright, surveying artists', copyright-owners', and users' points of view. Morreale et al. (2024) offer a post-Marxian perspective. While they focus on the unwitting labor of ordinary internet users (e.g., solving CAPTCHAs) rather than on creators' intellectual property, their concerns about the non-consensual use of data overlap with the issues discussed here. For journalistic reporting, see Chayka (2023).

[2] See Fuchs (2014) for a Marxian argument that tech companies problematically exploit online users by appropriating the value of their contributions for profit.

[3] See Shoemaker (2024, 13–14).

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on the act of copying itself. Even if we grant that AI programs do not contain copies of training data in any legally or morally relevant sense, the training process itself inevitably involves copying. I argue that creators possess a property right to restrict the *technologies* used to copy their work, thereby allowing them to block the use of web crawlers for AI training. On this view, AI training requires consent only if a creator explicitly invokes the right to block web crawlers. Importantly, this argument does not hinge on any claims about intellectual property. Rather, it hinges on non-intellectual property rights over *digital objects*.

Here is the plan for this paper. Section 2 covers preliminaries. Section 3 argues that even if creators have an extensive property right to restrict the use of their content for learning, it's regularly overridden by the human right to free thought. Section 4 argues that creators possess a property right to regulate the use of copying technology on their work. The paper concludes in Section 5 with a discussion of the policy implications of these moral arguments.

2 Preliminaries

2.1 The AI Training Pipeline

To contextualize the subsequent discussion, I'll begin with a description of how web content is harvested for generative AI. While the technical details provided here will not be exhaustive and do involve some simplification, they will be enough to clarify the matters at issue in this paper. I focus primarily on text-based models, as their operation is more straightforward to explain, though the same basic principles apply to image generators and other generative AI systems.

Such systems depend on massive training datasets assembled through automated web scraping. Web crawlers are automated programs that systematically visit web pages and retrieve raw content, such as HTML and text.[4] Unlike human browsing, where content is cached temporarily for immediate viewing, web crawlers store extracted content in centralized databases designed for indexing, analysis, and, in this case, AI training. This raw content undergoes substantial processing to create training datasets: the removal of redundant content, quality filtering, and format standardization. For text-based models, content is tokenized and chunked into training examples. The scale of these datasets is vast: for example, The Pile, used for language model training, aggregates 825GB of text from books, academic papers, and websites.[5] Even in Microsoft Word's rich formatting, that would amount to over 53 million pages of text. In plain text format, it would be over an order of magnitude more. If AI training involves theft, it is theft at a staggering scale.

With a massive dataset in hand, the model is ready for training. To do this, machine learning systems analyze the dataset to extract statistical patterns and encode them as numerical parameters in the resulting model.[6] Crucially, the final trained model

[4] For more information about this practice, see Olston & Najork (2010).

[5] For more information about The Pile, see Gao et al. (2020).

[6] For a detailed overview of LLM design, see Naveed et al. (2023)

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does not contain an archive of training examples. Rather, it encodes only statistical patterns extracted from those examples. Once trained, generative AI systems respond to user prompts by applying their learned statistical patterns to generate new content. For example, when a user asks GPT-5 to write a sonnet about artificial intelligence in the style of Shakespeare, the model generates novel output without accessing any database of training examples. That output may bear stylistic or structural resemblance to training data—the sonnet follows learned conventions of Shakespearean meter and rhyme—but it is not retrieved from or mechanically assembled from training examples. Rather, the generation process operates through the model's parameters alone.

While training image generators is more complicated, the same basic principles apply. As Shoemaker describes in more detail, systems like Stable Diffusion and DALL-E learn from millions of image-text pairs scraped from the web and generate novel images by applying learned patterns, not by retrieving training examples.[7] The details differ, but the basic structure is the same: creating these systems requires large-scale copying of web content, but the final models don't contain copies of that content.

With this characterization of AI training in hand, I am now in a position to characterize "learning," a notion that will be important going forward. First, it must be distinguished from mere copying. Copying reproduces data such that the copy can substitute for the original without significant loss of information, as in the case of photocopies. But copying alone doesn't constitute learning: nobody learns anything from the photocopies collecting dust in my attic. Rather, learning involves integrating information into a model in a way that supports inferences or actions that were previously less likely or impossible. While this may occur in a variety of ways, I will focus on the *extraction of patterns* from information. In this sense, both AIs and humans learn: AIs from their training data and humans from ordinary perception, testimony, etc.

2.2 Normative Framework and Scope

I will now clarify this paper's normative framework. Its arguments are firmly deontological, based on the rights of creators and learners. It very well may turn out that generative AI is a bane for humanity, or that requiring consent for AI training would result in a better world. That is not my concern here. Moreover, as I have already mentioned, this paper examines *moral* rights, not legal ones. While lawsuits pressing various lines against AI companies are currently making their way through the courts, they are orthogonal to the philosophical question of whether non-consensual AI training violates creators' moral rights.[8] Similarly, my focus is not on the norma-

[7] See Shoemaker (2024, 9–11) for an argument that generative AI programs don't contain copies of training data. Also see OpenAI (2021); Radford et al. (2021).

[8] For litigation on this topic, see *Andersen v. Stability AI Ltd.*, No. 3:23-cv-00201 (N.D. Cal. filed Jan. 13, 2023); *Getty Images (US), Inc. v. Stability AI, Inc.*, No. 1:23-cv-00135 (D. Del. filed Feb. 3, 2023); *Kadrey v. Meta Platforms, Inc.*, No. 3:23-cv-03417 (N.D. Cal. filed July 7, 2023); *Tremblay v. OpenAI, Inc.*, No. 3:23-cv-03223 (N.D. Cal. filed June 28, 2023); *Authors Guild v. OpenAI, Inc.*, No. 1:23-cv-08292 (S.D.N.Y. filed Sept. 19, 2023).

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tive foundations of intellectual property *law*. Even if Shoemaker is correct that they do not support a right to restrict AI training, moral property rights are not necessarily exhausted by the normative foundations of property law; the law must be sensitive to a variety of practical considerations (e.g., the need to incentivize innovation) and might thereby (reasonably) fail to enshrine certain moral property rights.[9] However, I will occasionally touch on the normative underpinnings of the law, insofar as they reflect deeper conceptual distinctions that bear on the scope of creators' moral rights.

Still, one might wonder why I am focusing on moral rights rather than legal ones.[10] Of course, legal rights will more directly shape the behavior of AI companies. However, these are still very much in flux, and it is worth asking what they *should* be. If creators have moral rights against the non-consensual use of their work for AI training, that gives us reason to enact legal protections for those moral rights. By contrast, if they lack such moral rights, legal restrictions may be difficult to justify on principled grounds. Additionally, the moral status of AI training is of independent theoretical interest, much as the moral status of meat-eating is worth examining regardless of the prospects for legal reform.

Next, two remarks about scope. First, whereas much of the existing discussion of non-consensual AI training has focused on art, my arguments are meant to apply to *any* content that may be used as training data: text, images, videos, etc.[11] Although some property rights may apply only to certain sorts of content, I aim to identify rights that apply to all media, whether in the form of news articles, artistic works, or even social media posts.[12] Second, I acknowledge that even if generative AI systems don't contain training data, they do sometimes reproduce copyrighted content as output, such as a verbatim *New York Times* article or a blatant image of the Nintendo character "Mario," without creators' consent.[13] While these cases are surely violations of intellectual property, they concern the *output* of generative AI. The matter of training concerns *input*.[14] To keep the discussion focused, I will for the sake of argument assume that the AI systems in question never violate copyright in the output sense.

2.3 The Dialectical Challenge

I am now ready to present the dialectical challenge facing those who would argue for restrictions on AI training. The first point to note is that granting narrow, copyright-like moral property rights to creators likely does *not* protect them from non-consen-

[9] See Shoemaker (2024) for a sustained argument that copyright law should not be modified to protect artists against non-consensual AI training.

[10] Thanks to an editor for pressing me on this point.

[11] Although Shoemaker (2024) focuses on image generators, many of his arguments extend to other cases of generative AI. The dialectical challenge he presents is thus quite broad.

[12] Fuchs (2014) likewise advances an argument of broad scope, contending that ordinary users' online activity—posting, commenting, browsing—constitutes labor that is exploited by big tech companies.

[13] For details, see *The New York Times Company v. Microsoft Corporation* et al., No. 1:23-cv-11195 (S.D.N.Y. filed Dec. 27, 2023).

[14] Of course, some inputs may constitute straightforward copyright violations, such as the use of pirated content as training data. While important in their own right, such cases fall outside this paper's scope.

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sual AI training.[15] As I've already mentioned, generative AI programs don't contain copies of training data. And while the training process itself does involve making copies, so does ordinary browsing. Indeed, whenever we view a webpage, the content is downloaded and temporarily stored (i.e., cached) on our devices. So, we need some way to normatively distinguish between copying by OpenAI's web crawlers, for instance, and copying for ordinary browsing.

A natural suggestion is to distinguish these forms of copying based on their *purpose*. But why should we think that copies made for AI training, unlike those made for browsing or indexing, require the creator's consent? There are two ways one might defend this claim: either it reflects a fundamental property right, or it is grounded in other, independently motivated property rights. The former approach strikes me as objectionably *ad hoc* unless the proposed fundamental right can be demonstrated as theoretically desirable for reasons that extend beyond the present discussion. Since I cannot think of a plausible way to do that, I will set the proposal aside.

So, we are on the hunt for independently motivated property rights that would justify protecting creators against the non-consensual use of their content for AI training. The major difficulty concerns how to accomplish this without entailing implausible moral restrictions on human learning. To illustrate, consider the suggestion that creators have an extensive, patent-like "right to exclude others from making, using, offering for sale, or selling" their work.[16] Shoemaker objects as follows:

> "The problem with this is that art is ubiquitously and perhaps necessarily 'used' by other artists as part of the creative process that produces new art. Virtually all art is produced via an artistic process that involves the artist observing and learning from art produced by others, and involves copying stylistic elements and motifs from these works. But if this is the case, if artists were given this expansive right to exclude, artworks that are inspired by or responding to other works of art would be required to licence the works that inspired them or be copyright infringing, in the same way that 'follow-on innovations' require the user to licence the patents that the follow-on innovation depends on to avoid patent infringement."[17]

If creators have such extensive property rights to control the use of their content, it seems that they must also have a right to block learning from it, whether by humans or AIs—extracting patterns is a "use," after all. But, of course, it would be absurd to suggest that creators possess moral rights restricting humans from learning from the content they produced. The problem, then, is that granting creators extensive property rights over use restricts not only AI learning, but also human learning. Even setting aside the question of law, such a moral conclusion is deeply implausible. And worse,

[15] For an extended argument to this effect, see Murray (2023).

[16] Although this language is drawn from the way patents are characterized in 35 U.S.C. § 154 (2006), I am not proposing to extend patent protection to expressive works. The phrase serves only to illustrate the breadth of the hypothetical property right under consideration.

[17] Shoemaker (2024, 13–14).

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since learning is quite literally the foundation of the creative process, the extensive property rights under consideration threaten to entirely neuter our creativity.

Learning from original works underpins a wide range of legitimate human pursuits. For instance, to imitate a work's style, one must extract the patterns that constitute its stylistic features, whatever those may be; I cannot paint *Sunny Day* in the style of *Starry Night* without first learning Van Gogh's style. Even less imitative works of inspiration involve learning from existing works. While I won't attempt to define "works of inspiration," I have in mind examples like Joyce's *Ulysses* (inspired by Homer's *Odyssey*) and Disney's *The Lion King* (inspired by Shakespeare's *Hamlet*). While such works retain important features of the originals, rather than merely imitating them, the works of inspiration build substantially on them. With that in mind, we must distinguish between "works of inspiration" and works that were created merely *as a result* of inspiration. For example, a writer may be inspired by *Slaughterhouse-Five* to write a satire of the war in Ukraine, but the resulting work is not a "work of inspiration" if its only connection to the original is that it satirizes a war. By contrast, *The Lion King* qualifies as a work of inspiration in virtue of its structural similarities to *Hamlet*. And of course, this could not have occurred without extracting patterns from *Hamlet*. Thus, restricting learning restricts works of inspiration, too.

To deal with this problem, a proponent of extensive creator property rights might suggest narrowing them such that they only restrict AI training, not human works of inspiration. But as stated, this proposal treats the right to restrict AI training as a fundamental property right without offering any independent motivation, rendering it objectionably *ad hoc*.[18]

Another initially appealing idea is to identify some crucial difference between human and AI *learning* that uniquely justifies restricting the latter.[19] Although my characterization of learning in terms of extracting patterns doesn't leave much room for this strategy, perhaps some other characterization would better facilitate this kind of argument. While I am open to such proposals, I am skeptical, because the problematic feature of AI learning would have to be unavailable to humans *in principle*. To illustrate, suppose we were worried about the speed and scale of AI learning. An initial problem with this idea is that it's not clear why speed and scale should be subject to moral restriction. But even worse, such restrictions would extend to super-talented humans or, perhaps more realistically, groups of such humans working together.[20] This would entail undue limitations on human group learning. Another possibility is that human learning is somehow *qualitatively* different and thereby worthy of unique moral protections. However, it's not remotely clear what the qualitative difference would amount to or why it should be normatively relevant. With that in mind, I set such proposals aside.

[18] See Shoemaker (2024, 15) for a similar argument.

[19] Thanks to an anonymous reviewer for pressing me on this point.

[20] There has been an enormous movement among epistemologists in recent years to characterize notions like group knowledge, understanding, and inquiry. See Habgood-Coote (2019, 2022); Boyd (2021); Khalifa & Goldberg (2025).

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Thus, the dialectical challenge remains to identify a property right that can restrict AI training while leaving human learning and inspiration intact, and to do so in a way that is independently motivated rather than arbitrarily tailored to block AI training.

3 Intellectual Property Rights and Freedom of Thought

3.1 Patent-like Property Rights

This subsection supposes that creators enjoy a patent-like, extensive *pro tanto* right to control use of their content. I argue that even if this is true, humans possess a stronger, countervailing right to freedom of thought that regularly overrides these *pro tanto* rights. Since AIs lack any right to freedom of thought, creators' *pro tanto* rights to restrict learning are left intact, enabling creators to restrict AI training.

I will not *argue* for the claim that creators have a *pro tanto* right to control use of their content. Rather, my goal is to show that, *contra* Shoemaker, the existence of such a right does not automatically intrude upon human works of inspiration as collateral damage. As I explain in Sect. 3.3, I suspect that a property right as broad ranging as the one supposed here will prove implausible for independent reasons. However, I hope that by addressing what I take to be among the most serious objections to this family of proposals, my argument will at least show that they merit further consideration.

Still, it's worth saying something to motivate the idea that creators possess *pro tanto*, patent-like property rights over the use of their work. The main reason is that copyright-like moral protections are probably not robust enough to restrict non-consensual AI training. Copyright primarily protects against the copying and distribution of expressive content, while allowing exceptions for activities like research and transformation. Because generative AI training often involves copying that could plausibly be viewed as informational or transformative, it's doubtful that copyright-like moral rights would reliably block such practices. By contrast, a more extensive property right over the *use* of one's work would give creators clearer grounds to object to their work being used for AI training without consent, even if the training process is transformative.[21] With this in mind, the majority of this Section will proceed under the (admittedly controversial) assumption that creators possess extensive, patent-like rights to restrict the use of their works.

Recall the dialectical challenge: to identify an independently motivated property right that restricts AI training while leaving intact human learning and works of inspiration. To address this, the key claim will be that a uniquely human right to freedom of thought may override creators' property rights to restrict the use of their content, thereby restricting AI training without impacting human creative activity. Among

[21] One might immediately object that patent-like rights to restrict the use of a work would also restrict the *consumption* of that work (e.g., reading, listening to a song, etc.). In reply, it's plausible that even if creators do possess patent-like moral rights, they forfeit the right to restrict consumption when they publish work online.

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the most plausible and primitive moral rights is freedom of thought. If we are not entitled to think what we want, it's hard to imagine what we might have the right to do. To lack control over what one believes, considers, or imagines would be to lack the very foundation of autonomy. Surely, then, the right to free thought is *extremely* powerful—far more powerful than a creator's property right to restrict the use of their content. This is why the reader is under no moral obligation to carefully reflect on this paper's arguments, much as I might wish to impose such a requirement. Put simply, creators cannot impose moral requirements on how consumers mentally engage with their work, because freedom of thought far outweighs creators' property rights, no matter how extensive they might be.

Since the extraction of patterns in humans is a cognitive process, the right to free thought protects human learning from even very extensive creator property rights. But why can't creators constrain the *production* of works that result from that learning? To keep the discussion grounded, I will focus on works of inspiration, but my subsequent remarks apply to production resulting from human learning more generally. While being inspired itself is a mental activity and thus protected by freedom of thought, the production of works of inspiration goes beyond thought and thus seems to remain within the scope of creators' extensive property rights, unduly restricting human creativity. More concretely, why should the right to free thought enable Disney to create *The Lion King* rather than merely plan it out? To deal with this problem, I will adapt Shiffrin's defense of free speech. Her general point is that free speech is essential to freedom of thought, so free thought protections extend to free speech. This bears on the current issue because speech, like creating works of inspiration, is an activity of production.

Shiffrin (2015) argues that free speech protections are (partially) grounded in free speech's role in thought and the development of the self *qua* thinker.[22] She identifies the following eight interests that agents have *qua* thinkers: to develop their capacity for thought, apprehend truth, exercise imagination, act as moral agents, become distinctive individuals, respond authentically, live among other autonomous agents, and receive appropriate recognition and moral treatment.[23] Crucially, free speech is essential for realizing these fundamental interests of thinkers; externalizing our thoughts through speech, writing, art, or music is not only a way to share what is already formed internally, but also a condition for fully forming, identifying, and evaluating our thoughts. For example, observe that the process of reasoning and the process of speech cannot be sharply distinguished. Indeed, I've changed my mind about the topic of this paper several times in the course of writing it. This phenomenon is ubiquitous, suggesting that freedom of speech is essential to freedom of thought. Therefore, insofar as we have a right to develop as thinkers, we have a right to free speech.

Put simply, humans have powerful rights to develop as thinkers, and because thinking is inseparable from speech, they have correspondingly powerful rights to free speech. Now observe that since speech is an activity of production, free thought protections extend to certain kinds of production. I propose that if free speech is

[22] See Shiffrin (2015, 80).

[23] Shiffrin (2015, 86–87).

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protected as essential to free thought, then so too should the creation of works of inspiration—they are also a means by which thought is developed, evaluated, and shared, perhaps even more so than ordinary speech. Indeed, creating a work of inspiration is often part and parcel of how one processes the original. Creators, then, cannot prohibit works of inspiration any more than they can tell you not to think about what they've created. Of course, creators may legitimately restrict *access* to their work (e.g., via paywalls); freedom of thought does not justify the theft of content on grounds that it may provoke a work of inspiration. However, once accessed, the freedom to think, including via the creation of works of inspiration, cannot be overridden by the original creators' property rights.

But even if the creation of works of inspiration is protected by freedom of thought, why should those who take inspiration be allowed to economically benefit? In reply, I argue that the original creator's *pro tanto* property right bears only on causally proximate uses of their work. The act of extracting patterns is causally close to the original work and thus could, in principle, fall within the domain of the creator's property rights, even if those rights are overridden by free thought considerations. By contrast, once inspiration has led to the creation of a new, transformative work, the causal chain between the original and the new becomes remote. At that point, the work of inspiration is sufficiently distant from the original that the original creator's *pro tanto* right over their work cannot reach the new work. The act of selling a work of inspiration, then, is not a "use" of the original work in any meaningful sense, because the normative relevance of the original creator's property rights has dissipated along the lengthy causal journey to the new work. Any economic benefit from the new work is thus incidental to the protected expressive activity of the new creator and does not infringe on the original's rights.

If the above arguments are successful, they show that even if creators have extensive property rights restricting the use of their works, humans have more powerful rights to free thought that protect works of inspiration and other creative uses of original works. By contrast, ChatGPT, for instance, has no moral status and thus no right to freedom of thought.[24] Indeed, it arguably cannot even "think," at least for now.[25] This blocks the analogy between AI training and human learning. Far-ranging creator property rights may thus provide a motivated theoretical foundation for restricting AI training without undue restrictions on human creativity.

If creators possess such extensive property rights, what sort of consent is required to use their work for AI training? Must AI companies explicitly seek permission to use content in training data, or need they only abide by any explicit restrictions a creator communicates? In general, extensive property rights require *explicit permission*. For example, we must ask permission before using a neighbor's lawnmower, even if it's left on the property line and there is no note indicating that we shouldn't. I see no reason why a creator's extensive property rights would be any different.

[24] For discussions of the relevance of consciousness to moral status, see Shepherd (2018); Lee (2025); Smithies (forthcoming). See Chalmers (2023) for a mostly skeptical discussion of LLM consciousness.

[25] For arguments to this effect, see Bender et al. (2021); Searle (1980).

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3.2 Objections

To prevent the free thought argument from unduly implying that creators have an obligation to grant access to their work, I distinguished *access* from *use*. The idea was that even if the human right to free thought overrides creators' rights to restrict use, it doesn't override their right to restrict access. One might object that this maneuver fails in the present context, because *online access itself requires copying* (i.e., caching). Presumably, copying qualifies as a "use," thereby falling under extensive, patent-like property rights. So, given the nature of the internet, a right to restrict use entails a right to restrict access, including that of humans. This threatens to render even the basic consumption of published online content morally problematic, the free thought argument notwithstanding.

In reply, I argue that by publishing work online, creators forfeit the right to restrict the minimal, technologically necessary copying that access entails. The essence of publication is to enable engagement, which in online contexts necessarily involves copying for display and temporary storage. Thus, even if creators initially possess extensive property rights, the act of publishing forfeits those related to human access. By contrast, creators haven't forfeited the right to restrict AI training, because online publication does not essentially involve enabling access by web crawlers—indeed, robots.txt files often prevent web crawlers from accessing published material. Of course, this should not be construed as suggesting that by publishing their content online, creators forfeit *all* their rights to restrict copying. Ordinary copyright infringement, such as republishing work on another site, is still restricted, because the act of initial publication doesn't require the production of such copies. By publishing, creators only forfeit their right to restrict the copying essential to the act of publication.

My objector might remain unmoved, because even if creators forfeit any right to restrict copying essential to publication, perhaps they nonetheless retain broad rights that unduly restrict human activity. Following this line of argument, suppose that creators begin with maximally broad rights to restrict use and only give up those constitutive of publication (i.e., caching). Among the rights they retain is the right to restrict access for *particular purposes*, including for the purpose of learning. This suggests that creators could block humans from permissibly caching (i.e., browsing) their work even for purposes like learning to draw. When taken to its logical conclusion, this position seems to permit just *one* sort of non-consensual consumption: *purposeless caching*. And perhaps passive viewing. *All* other activities, for *any* purposes, will be automatically restricted by creators' extensive property rights. In other words, the objector accuses my position of entailing that without special permission, we are confined to mindless scrolling. To avoid this implication, I must find a non-arbitrary way to exclude purpose-sensitive access restrictions from creators' extensive property rights.

To this end, I argue that sweeping, purpose-sensitive rights to restrict access are incoherent. The objection's premise is that the act of publication constitutively requires forfeiting the right to block caching but doesn't affect rights to block purpose-sensitive caching. But if it's coherent to publish while restricting access for the sake of learning to draw, then why shouldn't it *also* be coherent to publish while restricting access to those *without* any particular purpose (i.e., aimless scrollers)? It

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is easy to imagine trying to publish content with the proviso that only careful readers are morally permitted to access it. In fact, I can only think of one non-arbitrary way to determine which purpose-sensitive restrictions on access are compatible with the act of publication and which are not: publisher or creator intentions. This raises difficult follow-up questions: is it coherent to publish a work such that there is exactly one permissible reason to access it, or is that too restrictive? And what if the creator has no specific intentions in the first place? If these sorts of questions do not admit precise, non-arbitrary answers, then the whole notion of purpose-sensitive access restrictions collapses into incoherence, providing a principled reason to exclude them from creators' extensive property rights. We should conclude, then, that once content is published, any purpose-sensitive rights to restrict access to it are forfeited, and creators' property rights restrict only its post-access *use*.[26]

While identifying good answers to the above questions strikes me as hopeless, the objector may point to legal precedent for purpose-sensitive restrictions: under U.S. copyright law, whether the fair use doctrine permits or prohibits copying depends on the purpose of the use, with courts often distinguishing between personal use, scholarship, and commercial exploitation. For example, copying for transformative educational purposes may be allowed, while copying for commercial resale is not. Although copyright law of course does not restrict caching online art for learning, it suggests that purpose-sensitive restrictions on access can be made theoretically respectable.

This is a false hope. In the context of copyright, purpose-sensitive restrictions on access flow from restrictions on *use*; copying for commercial exploitation is prohibited because commercial exploitation of copyrighted works is prohibited. That's a non-arbitrary justification for purpose-sensitive access restrictions. But similar reasoning won't work for the extensive, patent-like creator rights under discussion here. Recall that creators' supposed right to restrict the post-access use of their work is typically overridden by freedom of thought. So, there is no normative source from which purpose-sensitive access restrictions may flow. Without that source, the objector must find other grounds for purpose-sensitive access restrictions. And as I already argued, that's a dubious proposition. Thus, granting creators extensive, patent-like rights to control use doesn't objectionably imply that they retain the right to restrict access for human learning.

3.3 Against the Supposition of Patent-like Creator Rights

My main goal in this Section has been to address Shoemaker's objection that granting extensive, patent-like protections to creators implies objectionable restrictions on human creativity. But even if my arguments have succeeded, they do *not* establish that creators possess such rights, because I began with that as an assumption. However, the assumption is dubious for independent reasons, even if it doesn't automatically entail undue restrictions on human works of inspiration.

[26] It might be argued that among creators' extensive rights includes the right to regulate which sorts of copying technology are used to access their works. Since I will address that argument separately in the next Section, I leave it to the side for now.

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Expressive works of the sort AI systems train on are legally protected only by copyright, and there are good reasons to think that the differences between copyright and patent protection are not merely arbitrary legal contrivances. For example, copyright protections do not restrict other creators from publishing overlapping content, so long as it's *independently* created. By contrast, patents *do* protect inventors against independent creation. Chatterjee argues that this distinction reflects a deeper difference in how artistic works and inventions are individuated: artistic works are "author-individuated," such that authorship is partly constitutive of the work's identity, while inventions are "structure-individuated," making their identity independent of who creates them.[27] This explains why patents can justifiably block others from independently creating the same invention, whereas copyright cannot block others from independently producing a similar work. If correct, this suggests that extending patent-like control over the use of expressive works would conflict with the nature of authorship, casting doubt on the assumption that creators possess such extensive rights. With this objection in mind, I'll now seek to ground restrictions on AI training in another way.

4 The Proprietor's Right to Restrict Copying Technology

The argument in this Section relies on an analogy between content-hosting websites and art galleries. This choice does not restrict my argument's scope to art; I could just as easily have picked bookstores, movie theaters, concert venues, comedy clubs, etc. The key point is that ownership or otherwise morally grounded control of such environments confers *proprietary rights*. Among these rights is the right to regulate the use of copying technologies. Thus, website proprietors have the right to block web crawlers from collecting their content while permitting the caching associated with ordinary browsing.

Before I begin, note that this argument will *not* rely on any claims about intellectual property. Indeed, everything I say about proprietary rights applies equally to furniture stores as to art galleries, even though furniture stores don't sell expressive works. While an article used for AI training undoubtedly contains expressive content, it is also instantiated on the website of publication as a *digital object*, with a particular functional and technical structure. An article, as an *intellectual object*, consists in its expressive content and meaning; as a digital object, it exists as a structured sequence of data embedded within a web infrastructure, subject to technical operations like storage, transmission, and parsing. Whereas intellectual objects are not individuated by their storage context—for example, the same token intellectual object can be realized on multiple servers—digital objects *are* individuated by their particular instantiation in a storage environment, which includes factors like location, format, metadata, and technical addressability.

Like other objects, digital objects are subject to their own property rights, distinct from intellectual property rights. For example, companies own and restrict access to

[27] See Chatterjee (2023) for an argument that expressive works and inventions are importantly different and warrant different intellectual property rights.

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compiled datasets, cloud-stored files, and virtual assets, even when they do not own the intellectual property behind the content. These rights attach to the digital objects as such, not the abstract works they embody. There are numerous other examples: non-fungible tokens (NFTs), cryptocurrencies, encrypted eBooks, and even Fortnite skins. Like other property rights, those governing digital objects may entail the right of exclusive access and transfer.[28] While the most intuitive strategy to restrict AI training begins with intellectual property, protections over digital objects represent a promising, underexplored alternative. A published article, for example, is just as much a digital object as it is an intellectual one, entitling owners to non-intellectual property protections. Thus, we may consider analogies to tangible property. To maximize parity, my analogy will concern tangible property that also happens to embody an expressive work.

Since consumers are less likely to purchase an artwork when a copy is available online, many galleries restrict photography. This is merely one instance of a proprietor's right to arbitrarily regulate behavior within their places of business. They are within their rights to remove a patron for sketching a copy but to permit a different patron to do so (as long as they are not motivated by discrimination against a protected class); they can permit iPhone photography but ban DSLR cameras; or they can entirely ban photography on the premises. It doesn't matter if the patron wants to use the photos to get better at drawing, stare at them before bed each night, or delete them the moment they get home. Generally speaking, proprietors may restrict which technologies patrons may use to copy works, if any. The same is true of bookstores, movie theaters, concert venues, comedy clubs, etc. So why shouldn't it be true online?

Like art galleries, most websites that publish content are *privately* owned. Just as galleries are private spaces open to the public under conditions set by the proprietor, websites grant *conditional access* to visitors while retaining the right to set terms of use and to impose technological restrictions. By contrast, subway drummers, for instance, have no moral right to regulate the recording of their music, because the subway is *public*. Indeed, that is among the chief reasons why many people perform there. Admittedly, there is a sense in which publishing work online feels similar to performing on the subway—anyone can visit a website, much as anyone can walk past a subway drummer. This might tempt us to think that by publishing online, creators forfeit their right to restrict how their work is copied. However, "easy to access" shouldn't be conflated with "public." A gallery may be easy to access, too, but that doesn't strip proprietors of their right to control copying technology. Similarly, site owners have the right to restrict web crawlers.

I've now established that websites are spaces where proprietary rights typically apply. But how could a proprietor's right to restrict copying technology apply online, given that ordinary web browsing inevitably involves caching?[29] The challenge, then, is to distinguish between copying by web crawlers and copying by ordinary browsing. To address this, observe that ordinary caching and web crawlers are *techno-*

[28] Of course, it is by now well-known that most digital "property" rights are better construed as licensure rights (Perzanowski & Schultz 2016). Still, many digital objects are sold outright.

[29] Thanks to Eric Shoemaker for pressing me on this point.

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logically distinguishable. Indeed, if they weren't distinguishable, then websites could not ever successfully block web crawlers while allowing ordinary browsing. Yet in practice, they do so (with mixed success) via robots.txt files and other access-control measures. Website proprietors may thus restrict OpenAI from using web crawlers while permitting ordinary caching, just as a gallery proprietor may restrict a patron from using a DSLR camera while permitting iPhone photography.

In the Introduction, I said I would argue that creators have the right to restrict copying technology, but the above argument only grants that right to site proprietors. But do those rights extend to creators? Before addressing this question, it's worth highlighting that whether or not the right to restrict copying technology begins and ends with site proprietors, AI companies must still respect it. Since my main aim in this paper is to provide normative foundations for restrictions on AI training, that is enough. Moreover, creators are sometimes able to exercise significant leverage over proprietors. For example, suppose a proprietor who generally permits any photography is told by an artist that "I will only permit you to display my piece under the condition that you don't let anyone photograph it with a DSLR, a policy clarified in a special note under the piece." Of course, the proprietor can refuse to display the piece. But if they think it will sell for a high price, they might agree to those conditions, in which case permitting such photography would wrong the artist. Thus, via negotiations with proprietors, creators may acquire protections against their work being copied by particular technologies.

Still, the core concern is that the right to restrict copying technology stems from the right to regulate the space in which copying occurs rather than direct rights over the object to be copied. And if creators don't control the space, then they don't control the use of copying technology. I am prepared to grant this point. However, I don't think it makes a difference to the normative status of most instances of copying, especially online. Creators can publish content in two broad contexts: platforms where they hold proprietary rights (e.g., their own site) and platforms where they lack such rights (e.g., someone else's site). When creators hold proprietary rights, they possess the right to regulate the use of copying technology. True, they don't possess it in virtue of having created the content, but why should they care where it comes from, so long as they possess the right? In the other case, when creators publish where they don't have proprietary rights, they must agree to the publishing site's terms anyway. Via their consent, whatever intrinsic rights to control copying technology that creators might have initially possessed are erased, overridden, or transferred to the site's proprietor. Thus, while my argument admittedly doesn't give creators the right to control copying technology on platforms they don't own, *no argument will*, because publishing on someone else's platform generally entails forfeiting that right. From a practical point of view, it makes no difference whether a creator forfeited their right or never had it in the first place—either way, they don't have the right. So, in any case, it makes no difference whether it is the creators or site proprietors who initially possess the right to restrict copying technology.

Note that the right to restrict copying technologies doesn't unduly restrict human creativity. To illustrate, suppose a patron with a photographic memory goes to a gallery without a camera and subsequently uses what they remember to improve their art. Since the patron didn't rely on a restricted copying technology, the proprietor has

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no legitimate complaint related to copying, even if the patron's memory is as flawless as a DSLR camera. Of course, it would've been within the proprietor's rights to simply refuse that patron access to the gallery. But once the patron is inside, freedom of thought protects their use of perception and memory.[30]

The right to restrict copying technology is far weaker than the extensive property right proposed in Section 3. Whereas that right requires companies to obtain permission to use content for AI training no matter what, galleries must explicitly communicate anti-copying policies for them to have force—they cannot enforce rules *post hoc*, such as by demanding that a patron delete a photo once it has been legitimately taken. Similarly, creators are wronged by AI companies only if their work is used as training data *despite a clearly communicated invocation* of their right to restrict web crawlers. The implication, I suspect, is that most AI training to date has been permissible, since very few creators and publishers instituted anti-AI training policies until quite recently.

5 Policy Recommendations

This paper has offered two independent foundations for normative restrictions on AI training. The first presupposes that creators enjoy far-reaching, *pro tanto* property rights to restrict the use of their content. However, it contends that humans, but not AI systems, possess a right to free thought that typically trumps creators' *pro tanto* rights. If sound, the argument requires AI companies to obtain permission before using work as training data. The second argument draws on an analogy to art galleries to show that site proprietors possess a right to control which sorts of technologies may be used to copy works on their site. The upshot: if proprietors clearly communicate that web crawlers may not be used to copy their work, then AI companies must obey.

The moral arguments developed here have direct policy implications. Many sites already provide instructions to web crawlers via robots.txt files. A natural suggestion is to implement a standardized way for web crawlers to determine which content they are permitted to copy, and to legally require entities in charge of these web crawlers to obey the commands therein. If the first argument succeeds, and creators have extensive moral property rights, then even sites without such a file must be passed over. If only the second argument succeeds, then web crawlers may mine such sites to their digital hearts' content. In either case, any explicit restrictions must be respected.

One difficulty with this recommendation concerns search engines. Suppose that only my second argument succeeds, and creators just have the right to restrict copying technology, not what's done with those copies. Search engines like Google use web crawlers for indexing, and few sites can attract enough traffic to survive without subjecting themselves to it. But those web crawlers may also, concurrently, be used to gather training data. If site owners cannot legitimately block only one use of the web crawler, then they are out of luck and must succumb to AI training.

In reply, note that gallery owners may legitimately permit photography while simultaneously demanding that photos be deleted before leaving the premises, or

[30] Note that this invocation of free thought doesn't rely on the idea that creation is intertwined with thought and is thus entirely independent of the argument from Section 3.

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after just 30 seconds. While these would admittedly be strange (and difficult to enforce) policies, they still fall under the moral rights of proprietors. Similarly, site owners may demand that any copies made by web crawlers be deleted after a short period of time. To enforce this right, we might implement policies according to which web crawlers must obey commands about how long the scraped data can be stored, thereby permitting relatively faster indexing while stopping slower AI training.

Of course, these proposals only scratch the surface, and it remains uncertain how they would interact with existing regulatory frameworks across different jurisdictions. Policymakers will also have to consider how to balance the moral property rights discussed here with the need to incentivize innovation.[31] Regardless, if the arguments in this paper have succeeded, then they should take the moral stakes of AI training more seriously. The rights of creators—to control how their work is used and copied—do not vanish in the face of technological convenience.

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[31] For a defense of the non-consensual use of copyrighted material for AI training on innovation grounds, see Lemley & Casey (2020).

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